

RAGHAVENDRA DINESH

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SUMMARY

Graduate Robotics and Artificial Intelligence with 3 years of work experience at Bosch in Data Science and Automation. Competency in problem-solving, communication, and teamwork with relevant experience in Generative AI, Computer Vision, and NLP.

PROFESSIONAL EXPERIENCE

Multi Modal Deep Learning Researcher

June 2024 - Present

Arizona State University

Plant Image based Health Diagnostics System with BLIP-2 Image Captioning using Vicuna 7B and Gemini API

- Achieved 98% classification accuracy on PlantVillage Dataset with ResNet50 classifier quantized using Onnx and TFLite.
- Obtained Rouge-N score of 0.63 on text generated from Gemini API passing prompts based on image classification output.
- Trained Google's ViT for same image dataset with 43% accuracy on 50 epochs. Working on SWIN for Low Level feature extraction.
- Tested Google Flan T5 and Vicuna 7B LLMs and observed higher Rouge-N score with Vicuna. Working on integration into BLIP-2.

Automation Engineer

Sep 2019 - Aug 2022

Bosch Global Software Technologies

Team Leadership

- Led a team of three engineers to develop tools enabling analysis of test results and report generation reducing time by 40%.
- Spearheaded the development of an Azure CAN Dashboard, that enables data transfer from ECUs to Azure Cloud for CAR VR.

RF Emission Analysis and Harmonic Anomaly Detection for Automotive Conformance

- Deployed a centralized React web app to view and store RF Emission data for conformance analysis using Django.
- Attained 85% accuracy in anomaly detection for RF Harmonics using BiLSTM Autoencoder.
- Saved 90% of time by generating plots and reports automatically.

Smart Inventory Tracking Using Asset Detection and OCR

- Led implementation on Raspberry Pi 3B+ using YoLov4-Tiny and Tesseract for asset detection and labeling saving 25 hours/week
- Developed Django app, automating asset labeling and notifications, improved asset accountability, reducing misplacement by 80%.

Artificial Intelligence Intern

Jan 2019 - Mar 2019

ConnectM Technology Solutions

Truck Driver Assistance System with Safety Constraints using Facial Recognition and Voice Assistance ChatBot.

- Built voice recognition and face recognition systems using CMU Pocket Sphinx and Haar Cascade and AWS Face Rekognition.
- Trained drowsiness detection using Facial Landmark detection and made sure modules can be deployed on Edge devices.
- Seamlessly integrated the modules onto a Raspberry Pi 3B, ensuring optimal performance, user experience, and road safety.

EDUCATION

Masters in Artificial Intelligence (Robotics and Autonomous Systems)

Arizona State University

3.97 GPA

Bachelor of Engineering in Electronics and Communication Engineering

PES Institute of Technology

3.20 GPA

Relevant Coursework: Artificial Intelligence, Linear Algebra, Modeling and Control of Robots, Statistical Machine Learning, Mathematical Foundations of ML, Perception in Robots, Embedded ML, Biomedical Image Processing, Advanced Computer Architecture.

TECHNICAL SKILLS

Programming & Scripting: Python, C & Embedded C, MATLAB/Simulink, C++, Java, JavaScript, HTML, Bash

Interests: Computer Vision, Deep Learning, Natural Language Processing, Generative AI, Transformers, Large Language Models

Tools and Platforms: Pandas, NumPy, TensorFlow, Keras, PyTorch, OpenCV2, Scikit-learn, TensorRT, Onnx, TFLite, SQL, GIT, Docker

Certifications: Machine Learning and Deep Learning Specialization by Andrew Ng, DeepLearning.AI TensorFlow Developer Specialization by Laurence Moroney, MLOps | Machine Learning Operations Specialization by Duke University.

PROJECTS

Negative CLIPort: Improved Robotic Arm Control enhanced with Negative Language Goals | Arizona State University Spring 2023

- Improved CLIPort to handle a wider range of requirements by making it more robust to negative language goals.
- Training data augmented with goals that specify actions arm avoids, and loss function modified to penalize conflicting actions.

Medical Image Analysis for Lung Disease Classification and Detection | Arizona State University

Fall 2023

- Leveraged models/techniques like Swin Transformer, Faster R-CNN, UPerNet, and U-Net for classification, localization, and segmentation of Chest X Rays like VinBigData CXR, SIIM-ACR Pneumothorax, MedMNISTv2 and MINIJSRT Datasets
- Developed a multi-task model to accurately analyze X-rays, achieving high performance in metrics like AUC, IoU, FROC.

Real-Time Hand Gesture Recognition System for Conference Calling and Media Control | Arizona State University

Fall 2023

- Utilized Google's MediaPipe framework to extract hand features from the image. Based on the app we developed 3 classifiers.
- Deployed ASL, conference call and media control apps on Raspberry Pi 4B. Achieved a high frame rate and 93% accuracy.

Facial Image Synthesis from Segmentation Maps and Sketches using GANs for Forensics | Arizona State University

Fall 2023

- Utilized CelebAMask-HQ dataset with Pixel2Style2Pixel GAN to transform sketches and segmentation maps to real images.
- The real face images generated with good accuracy and clarity and is useful for Detective work.

GAN Abstract Art fusion with LSTM Poetry for NFT Auctioning | Personal

March 2021

- Generated Artwork with poetry using GAN-LSTM and Stable Diffusion. Observed better results with GAN-LSTM.
- The integrated Art Poetry output images were auctioned as NFT.

EXTRA CURRICULUM

- Tutored at Salvation Army Kroc Center, teaching Reading, Writing, and Math for 2 years.
- Volunteered as weekend teacher 9 months for underprivileged immigrant kids with FSL India and Bosch.